# Anton Liutin

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## EDUCATION

**PhD, Agricultural & Applied Economics** 2021 – Expected May 2027
University of Wisconsin–Madison — dissertator. Advisor: Paul Castañeda Dower

**MA, Economics** 2020
New Economic School, Moscow, Russia
Thesis: "Polygamy and Inequality in a Male-Dominant Society" (advisor: Hosny Zoabi)

**BS, Theoretical Physics** 2018
National Research Nuclear University MEPhI (Moscow Engineering Physics Institute), Moscow, Russia

## RESEARCH INTERESTS

**Fields:** Water and natural-resource economics · development economics · institutional and political economy · behavioral and experimental economics · economic history

**Methods:** Machine learning and text-as-data · spatial econometrics and remote sensing · network and graph methods · field and framed experiments

## JOB MARKET PAPER

**"Who Runs Short in a Drought? Water Rights and Canal Delivery in Arizona"**
Full working draft. Links 54,946 Arizona fields to a directed canal network over 2016–2023. Across Arizona, even holding each field's water rights, well access, crop, and local weather fixed, fields with more farmed land upstream show larger falls in crop yield during sustained drought; farmed land downstream shows no such link. The gap is concentrated in junior-rights irrigation districts, which lose surface water first, and is absent in the drier 2010–2014 drought, when major canal deliveries largely held. Monthly delivery models of these canal systems predict where along the canal the water runs out; in those months, water use on fields past that point falls by about 3% to 7% when they have no well. Latest version: antonliutin.com/jmp.pdf.

## WORKING PAPERS

**"Behavioral Games and Water-Saving Technology Adoption in Uzbekistan"** (with Jamshid Shukurullaev)
Working draft. Framed field experiment with 190 farmers across four regions, drawing participants from the IFPRI farmer survey; conducted with CGIAR SPIA support. Latest version: antonliutin.com/behavioral-games.pdf.

**"Double Vision: Donor Information, Farmer Voice, and Irrigation Policy Reform in Uzbekistan"** (with Muzna Alvi, Paul Castañeda Dower, and Claudia Ringler)
Sets about 330,000 farmer Telegram messages against media coverage, donor documents, and legislation to ask whose concerns the law follows. Four state newspapers OCR'd in house; six channels classified in two languages by an ensemble of language models, checked against an expert-coded sample. IFPRI collaboration. Latest version: antonliutin.com/double-vision.pdf.

**"Water Conservation Through Simplified Drip Irrigation Scheduling for Cotton Cultivation in Uzbekistan"** (Jamshid Shukurullaev, Ahmad Hamidov, and **Anton Liutin**)
Complete manuscript. Three-season paired field validation plus a 32-farmer adoption panel (difference-in-differences).

## WORK IN PROGRESS

**"Impact Evaluation of Drip Irrigation in Uzbekistan"** (with Paul Castañeda Dower, Muzna Alvi, and Claudia Ringler)
Survey-based impact evaluation of drip-irrigation adoption using the 2024 farmer survey; IFPRI / CGIAR NEXUS Gains collaboration.

**Community testimony and environmental change in Alaska** (with Paul Castañeda Dower)
Language-model analysis of over a million turns of hearing testimony and regulatory comment from Alaska communities, paired with satellite measurements of environmental change.

## PRESENTATIONS

**New Economic School Brown Bag Seminar (online)** April 2026
"From Position to Exposure: Drought Amplification in Directed Water Networks"

**Central Asian Economics Conference, Astana, Kazakhstan** August 2025
"Using Behavioral Games to Understand Water-Saving Technology Adoption in Uzbekistan"

**CGIAR SPIA Webinar (online)** August 2024
"Behavioral Games to Explore Water-Saving Technology Adoption in Uzbekistan"

**NEXUS Gains / WEFE Central Asia Summer School, Tashkent, Uzbekistan** August 2023
"Water-Saving Technologies in Uzbekistan"

**Departmental seminars, University of Wisconsin–Madison** 2022 – 2026
Regular presenter: DevLab, Environment and Resource Economics Seminar, and Student Research Colloquium

## RESEARCH EXPERIENCE

**Research Assistant, Paul Castañeda Dower** Summer 2026
University of Wisconsin–Madison — community institutions and adaptation in Alaska: language-model analysis of hearing testimony and regulatory comment, paired with satellite measurements of environmental change

**Research Assistant, Dustin Frye and Gisella Kagy** Fall 2024 – Summer 2025
University of Wisconsin–Madison — environmental health and economic history

**Research Assistant, Paul Castañeda Dower** Summer 2023
University of Wisconsin–Madison — Russian economic history

**Research Assistant, Corbett Grainger** Summer 2022 – Spring 2023
University of Wisconsin–Madison — environmental and natural-resource economics

**Research Assistant, Andrei Markevich** Winter 2020 – Summer 2021
New Economic School — Russian economic history

**Junior Researcher, Hosny Zoabi** Winter 2020 – Summer 2021
New Economic School, Research Department — family and growth economics

**Junior Researcher** Winter 2017 – Summer 2018
P.N. Lebedev Physical Institute, Russian Academy of Sciences

## FIELDWORK

**Uzbekistan — framed field experiment on water-saving technology adoption** Nov – Dec 2024
Designed and ran a lab-in-the-field experiment with 190 farmers across four regions, with Jamshid Shukurullaev; supported by CGIAR SPIA.

**Uzbekistan — large-scale farmer survey (IFPRI / CGIAR NEXUS Gains)** Sep – Nov 2024
Prepared and fielded a survey of small and large farmers across five regions; trained and monitored local enumerators.

**Uzbekistan — qualitative fieldwork on water-saving technologies** Summer 2023
Conducted focus-group discussions and in-depth farmer interviews across seven regions on the adoption of water-saving technologies, with Paul Castañeda Dower in collaboration with IFPRI.

## TEACHING EXPERIENCE

**Teaching Assistant, AAE 636: Applied Econometric Analysis I** Fall 2026
University of Wisconsin–Madison — graduate applied econometrics

**Co-Instructor, AAE Math Camp** Summer 2026
University of Wisconsin–Madison — calculus and probability for incoming graduate students

**Teaching Assistant, AAE 350: World Hunger and Malnutrition** Spring 2024
University of Wisconsin–Madison — led four weekly discussion sections (cross-listed with Nutritional Sciences)

**Teaching Assistant, AAE 215: Introduction to Agricultural and Applied Economics** Fall 2023
University of Wisconsin–Madison — undergraduate introduction to agricultural economics

**Teaching Assistant, Development Economics (MA)** Fall 2020
New Economic School — course by Michele Valsecchi

**Private Mathematics Tutor** 2017 – 2022
Prepared students for mathematics-intensive economics BA/MA programs (New Economic School, HSE).

## AWARDS AND FELLOWSHIPS

**University Fellowship, UW–Madison** 2021–22, 2025–26
Nine-month university fellowship supporting first-year doctoral study (2021–22) and fifth-year dissertation research (2025–26).

**Deborah and David Penn Fund Research Presentation Award, UW–Madison ($1,000)** 2025
AAE Graduate Committee award supporting presentation at the Central Asian Economics Conference, Astana.

**CGIAR–SPIA Fellowship ($25,500)** 2023
Research fellowship awarded following external peer review; supported the behavioral-games field experiment in Uzbekistan (NEXUS Gains collaboration).

**CREECA Graduate Student Short-Term Research Award, UW–Madison ($3,000)** 2023
Competitive award for the project "Current Cooperation Problems under Common-Pool Resources in Uzbekistan."

## RESEARCH GRANTS AND PROJECTS

**CGIAR GENDER Impact Platform, WEAI-Climate Learning Grant ($100,000)** 2023–24
"Unveiling the Nexus: Climate-Smart Irrigation and Women's Empowerment in Uzbekistan." Named collaborator (survey design, fieldwork, analysis); PIs Muzna Alvi and Claudia Ringler.

**SPIA Supplemental Funding, IFPRI / NEXUS Gains team ($40,950)** 2024
Team funding for completion of survey, fieldwork, and outreach for the Uzbekistan research program; research-team member.

**SPIA Scoping and Travel Grant ($9,300)** 2023
Joint grant for scoping and preparatory fieldwork in Uzbekistan; co-applicant with Paul Castañeda Dower.

## SKILLS AND LANGUAGES

**Programming & software:** Python (pandas, GeoPandas, NumPy, SciPy), Stata, R, ArcGIS, QGIS, LaTeX, Git

**Machine learning & geospatial:** Google Earth Engine (Sentinel-2, Landsat, OpenET); satellite and crop-sequence classification; directed-graph construction and validation at national scale

**Text as data:** LLM classification pipelines — in-house OCR (DocLayout-YOLO, Surya; Cyrillic), open-weight model ensembles with majority voting, validation against hand-coded gold standards

**Languages:** Russian (native), English (fluent)

## REFERENCES

Paul Castañeda Dower (advisor), Associate Professor, Agricultural & Applied Economics, UW–Madison — pdower@wisc.edu
Corbett Grainger, Professor, Agricultural & Applied Economics, UW–Madison — corbett.grainger@wisc.edu
Daniel Phaneuf, Henry C. Taylor Professor, Agricultural & Applied Economics, UW–Madison — dphaneuf@wisc.edu